

Project Name : Gopher Industries Food Remedy API

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Ticket No. : RE029

1. Introduction

The purpose of this review is to identify the main limitations in the current Food Remedy database and explain how those limitations may affect what users see in the application. The current database documentation describes the product data as usable but incomplete. Missing ingredients, nutrient values, categories, brands, images, allergen information, and product coverage can affect scores, recommendations, badges, and scan results [1].

This report translates those technical limitations into simple user-facing messages and recommends where those messages should appear in the app.

2. Main Database Limitations

The current Food Remedy database has several known data-quality limitations. Sparse ingredient and nutrient data can reduce or remove diet tags, health tags, ingredient-risk badges, and health scores. Missing or weak categories can also reduce the quality of similar-product and healthier-product recommendations [1].

Product coverage is another limitation. The seeded Firestore dataset does not represent complete Australian retail coverage, so a valid barcode may still return no product if that item was never seeded. The source data can also become stale, contain noisy labels, or use ambiguous categories [1].

3. Measured Data Gaps

The DB038 documentation includes measured results from a 5,000-product sample. Before remediation, 4,523 records had missing or empty categories, 1,162 had missing or empty nutrient information, 1,085 were missing a brand, and 7 were missing a product name [1].

After remediation, integration checks passed, but 4,651 of the 5,000 records were placed in the broad “other” category. This shows that a dataset can pass technical validation while still having weak category detail that limits recommendation quality [1].

4. Database Limitation to User Impact Mapping

Database limitation	Possible user impact	Suggested user-facing message
Missing ingredient information	Diet, allergen, or ingredient-based tags may be incomplete	Some ingredient information is unavailable for this product.
Missing nutrition information	A reliable health score may not be available	There isn't enough nutrition information to calculate a reliable health score.
Product not found in seeded database	A valid scan may return no product	This product is not currently available in our database.

Missing brand, image, or label	The product page may look incomplete	Some product details are currently unavailable.
Weak or missing category data	Similar-product recommendations may be limited	We couldn't find enough reliable product matches for recommendations.
Stale source data	App details may differ from current packaging	Product information may differ from the latest packaging. Please check the label.
Sparse allergen data	Allergen detection may be incomplete	Allergen information could not be fully verified.
Heuristic allergen detection	False positives or false negatives are possible	Please check the product label for the most current allergen information.
Limited Australian coverage	Some valid products may not be recognised	This product may not yet be included in the Food Remedy database.
Weak recommendation peer data	Few or no healthier alternatives may appear	We couldn't find enough suitable alternatives for this product.

5. Nutrition and Health Score Limitations

Food Remedy requires a minimum amount of nutrition information before a full health score can be calculated. The current nutrition enrichment logic requires at least three known nutrients from the supported nutrient set. When there is not enough data, the full score remains unavailable and only limited tags may be generated [1].

For users, the app should avoid showing a provisional value as if it were a fully reliable health score. A clearer message would be: "There isn't enough nutrition information available to calculate a complete health score."

6. Allergen Information Limitations

Allergen detection is one of the areas where Food Remedy should avoid absolute claims. The current process uses keywords, regular expressions, and negation patterns. If ingredient text is missing or incomplete, the system may rely on existing source allergen tags, and unusual wording may not always be detected. DB038 explicitly describes the allergen output as heuristic rather than laboratory analysis [1].

The app should therefore avoid statements such as "This product is completely allergen free." A safer message would be: "Allergen information could not be fully verified. Please check the product label."

7. Recommendation Limitations

Recommendation quality depends heavily on category and nutrition data. Very broad categories such as "other", missing nutrient overlap, or flat health scores can result in weak matches or no healthier alternatives [1].

Instead of making the feature appear broken, Food Remedy could show: "We couldn't find enough suitable alternatives based on the available product data."

8. Where User-Facing Messages Should Appear

- After barcode scans: show the database coverage message immediately if the product is not found.

- On product detail screens: place missing ingredient, nutrition, brand, or image messages beside the affected section.
- In the allergen section: show uncertainty prominently rather than hiding it in general information.
- In recommendations: show a short explanation where healthier or similar products would normally appear.
- For health scores: show an unavailable or incomplete-data message instead of a misleading score.

The wording should stay short and non-technical so users do not need to understand Firestore, enrichment pipelines, or source datasets.

9. Claims Food Remedy Should Avoid or Qualify

Food Remedy should avoid making claims that the available data cannot fully support.

- “This product is completely safe for your allergies.”
- “This product contains no allergens.”
- “This is definitely the healthiest option.”
- “This product information is fully up to date.”
- “All Australian products are available in the database.”
- “No recommendation means there are no better alternatives.”
- “A successful database validation means every field is accurate.”

More appropriate wording includes phrases such as “Based on the available data...”, “Some product information may be incomplete”, and “Please check the latest product packaging.”

10. Recommendations

- Clearly distinguish missing data from confirmed negative information.
- Show simple fallback messages when brand, image, category, nutrition, or ingredient data is unavailable.
- Avoid displaying provisional health scores as final user-facing scores.
- Treat unknown allergen information as uncertain rather than safe.
- Show a clear message when a barcode is not available in the seeded database.
- Explain when recommendations are limited because of category or nutrition gaps.
- Inform users when product information may be outdated compared with current packaging.
- Keep user-facing wording short and avoid technical database terminology.
- Place limitation messages directly beside the affected feature.
- Continue monitoring source-data quality as coverage improves.

11. Conclusion

The Food Remedy database is suitable for product lookup, scoring, tagging, and recommendations, but the data is not complete enough to support absolute claims in every situation. The main limitations include missing nutrition and ingredient data, incomplete categories, limited barcode coverage, stale source information, uncertain allergen detection, and weak recommendation matches [1].

For users, the most important improvement is transparency. When data is incomplete, the app should explain that clearly rather than presenting uncertain information as confirmed. Simple messages such as “Some

product information is unavailable,” “Allergen information could not be fully verified,” or “This product is not currently available in our database” provide a clearer and safer user experience.

References

[1] Food Remedy API Team, “DB038 – Source Data Gaps and Limitations,” Food-Remedy_API repository, 2026.

[2] Food Remedy API Team, “API Documentation,” Food-Remedy_API repository, 2026.

[3] Food Remedy API Team, “DB032 Validation Runbook,” Food-Remedy_API repository, 2026.

[4] Open Food Facts, “Open Food Facts – Open Database of Food Products,” [Online]. Available: <https://world.openfoodfacts.org/>.